HF-T2CR: High-Fidelity Thin and Thick Cloud Removal in Optical Satellite Images Through SAR Fusion

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***Abstract*— Cloud removal can effectively address cloud contamination in optical remote sensing images. But the simultaneous removal of both thin and thick clouds remains a significant challenge due to their distinct features. Thin cloud regions permit limited observation of ground information, in contrast to thick cloud, which completely obscures such information, necessitating different treatment processes. While some previous methods could effectively handle either thin or thick clouds, they often resulted in a slightly blurry outcome for the other type. In this article, we propose a cloud removal scheme that handles thin and thick clouds as a cohesive entity. Specifically, we initially use a network based on residual architecture to recover optical information under thin cloud and generate a grayscale cloud mask. Through the grayscale mask and a predefined threshold, thick cloud regions are identified. Then, an encoder–decoder network is used to estimate the information of thick cloud regions labeled by the predicted mask. Next, synthetic aperture radar (SAR) images are used as auxiliary information for cloud removal, providing the most indicative features for the edges of cloud regions. Finally, a contextual feature transfer mechanism (CFTM) imports features from remote spatial locations to fill in thick cloud regions, enhancing both visual and semantic coherence. As a result, our approach resolves the spectral defects of fuzziness and incomplete cloud removal. Experiments conducted on the SEN12MS-CR dataset confirm that our method outperforms others in all metrics, including mean absolute error (MAE), spectral angle mapper (SAM), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM).**

***Index Terms*— Cloud removal, contextual feature transfer, optical image, synthetic aperture radar (SAR).**

I. Introduction

In recent years, remote sensing technology has significantly progressed, leading to the widespread use of optical remote sensing images across various fields, such as change detection [1], disaster monitoring [2], hyperspectral image classification [3], [4], and crop monitoring [5]. However, continuous multiyear observations conducted by the Moderate Resolution Imaging Spectroradiometer (MODIS) aboard the satellites Terra and Aqua revealed that clouds cover roughly 67% of the Earth's surface [6]. Therefore, optical images are susceptible to losing optical information caused by cloud contamination. Consequently, capturing clear optical images at specific time or continuously at regular intervals can be challenging. The presence of clouds can lead to the missing of optical image data, reducing the value of optical images for applications. This can hinder the scientific research and engineering applications of remote sensing data. For example, change detection [7], [8] involves comparing clear images from two different dates to detect land changes. However, clouds have a significant impact on change detection as they can affect the reflectance of spectral bands, potentially leading to the identification of nonexistent changes. Another example is the crop growth analysis [9] which may be hampered by cloud cover, resulting in data gaps. To accurately calculate crop growth-related indices, it is necessary to restore the missing data. Therefore, cloud removal is an essential preprocessing step for optical remote sensing image applications.

Clouds are categorized into thin cloud and thick cloud, depending on their level of opacity. The observable ground objects in thin cloud regions can provide optical information, although not very clear, which still could serve as auxiliary information. But there is no such information in thick cloud regions.

Previous work has done considerable research [10], [11] on single-image cloud removal, but generally only for a specific case in either thin or thick cloud, as shown in Fig. 1(a). In general, thin cloud removal is achieved by image translation techniques, which learns the map between thin cloudy images and cloud-free images using a neural network. Thick cloud removal typically uses image inpainting techniques, which mask cloud regions as invalid information, thereby eliminating the impact of cloud on the results. However, the techniques excel only with one type of clouds but falter with the other, which leads to the unsatisfactory results in treating the contaminated images since both the thin cloud and thick cloud usually co-occur in cloud-contaminated images. Therefore, some researchers tried to remove thin cloud first and then thick cloud using image translation and image inpainting in one pipeline. For example, Zheng et al. [12] used U-Net initially to extract and remove thin cloud, followed by a GAN to remove thick cloud. However, these methods are constrained

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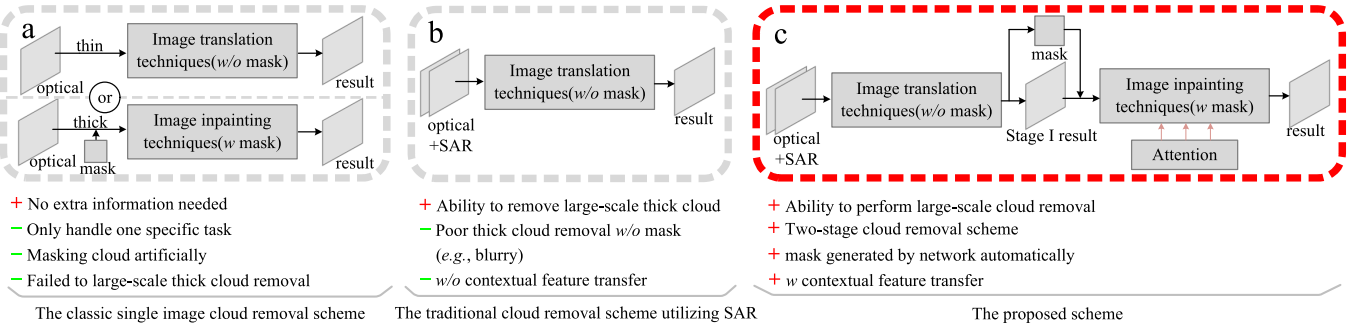


Fig. 1. Demonstration of the differences between the proposed scheme and other cloud removal scheme. We focus only on deep-learning-based cloud removal models here. (a) Single optical image cloud removal method. (b) Cloud removal methods that use SAR as auxiliary information, which map SAR image and cloudy image into cloud-free image. (c) Our novel scheme, which first removes thin cloud and identifies thick cloud regions; it then reconstructs these regions and leverages attention mechanism to fully exploit contextual spatial features. We have summarized the advantages (+) and disadvantages (–) below each subfigure.

to addressing thin cloud or small area of thick cloud. They are unable to effectively recover information beneath extensive area of thick cloud. In addition, they necessitate the provision of extra cloud masks.

The removal of large-scale thick cloud is a big challenge because there is no optical information in thick cloud regions. Some studies [13], [14] have used synthetic aperture radar (SAR) [15] data as auxiliary information for the reconstruction of information under cloud, as shown in Fig. 1(b). SAR is an active microwave sensor emitting microwaves that can penetrate the cloud. Therefore, information collected by SAR is not affected by clouds. Although SAR reveals no local detail comparing with optical images, it can describe the structural information on the ground. Image structure is the most indicative feature that plays a significant role in image analysis and image understanding [16]. Many image restoration researches [17], [18] involve first roughly drawing a structure to fill in the missing areas, and then proceeding with filling in the details. Therefore, it is appropriate to use the structural information provided by SAR for the reconstruction of thick cloud regions. Previous studies adopted image translation techniques to directly map SAR images and cloudy images into cloud-free images. However, these approaches introduce cloud-contaminated information into the network, resulting in outcomes that are prone to cloud interference, particularly for thick cloud, leading to incomplete or blurred cloud removal.

To address problems mentioned above, we propose a cloud removal scheme which divides cloud removal task into two subtasks: recovery of optical information beneath thin cloud and reconstruction of thick cloud regions, as shown in Fig. 1(c). This novel cloud removal approach enables the utilization of optical information beneath thin cloud while effectively eliminating the influence of thick cloud on the results. Initially, a thin cloud removal network based on the residual architecture [19] is proposed to remove thin cloud and generate a thick cloud mask. The skip connections within the thin cloud removal network act as a conduit for transferring optical information under thin cloud to the output of the network. However, this conduit is not optimal for handling thick cloud and can result in blurring in thick cloud regions. Therefore, reprocessing is required for the thick cloud regions in the recovered result. Second, a thick cloud removal network based on an encoder–decoder architecture is used to reconstruct the thick cloud regions marked by the mask generated during the thin cloud removal. Because of the self-similarity between lands in optical remote sensing image, information from other locations can be used to fill thick cloud regions. Inspired by contextual attention [18], [20], [21], [22], we use a contextual feature transfer mechanism (CFTM) to fill the thick cloud region with information borrowed from distant spatial locations to generate more fine-grained texture details and maintain the consistency of the global structure. CFTM can search for patches of spatial feature in the global scope and assign a weight to each patch. The weight indicates the importance of the patch. The greater the weight, the higher the similarity between this patch and the patch in the cloud regions. CFTM attaches more attention to these high-weight patches when filling cloud regions. In addition, the scheme uses SAR images as auxiliary information to assist in cloud removal, allowing the network to recover information for cloud regions based on the SAR images as a priori.

Our method considers the features of different cloud types in complex cloud scenes. First, it uses residual architecture and long skip connection to minimize variations in cloud-free regions, and the optical information under the thin cloud regions is preserved as much as possible. Then, the reprocessing of thick cloud regions using an encoder–decoder network tackles a critical limitation of previous methods in removing thick cloud incompletely, and the introduction of SAR images provides reliable ground information for removing large-scale thick cloud. Furthermore, our method has the capability to automatically predict cloud mask, reducing the cost associated with acquiring cloud mask data. In addition, we introduce the CFTM to address the issue of inconsistent blurry textures between cloud regions and surrounding regions in the results of previous methods, thereby enhancing the visual and semantic coherence of cloud removal results and enabling the model to generate cloud removal results with high fidelity. In CFTM, unlike other fields, we integrate the feature of SAR images to compute the affinity between patches in cloud-free regions and patches in cloud regions. We compute the affinity with higher confidence compared with other fields.

The main contributions of this work are as follows.

1) We design a novel cloud removal scheme of removing thin cloud first and then thick cloud as a cohesive entity. According to the distinct features of the two regions,

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we adopt image translation techniques to restore information within thin cloud regions, enabling the utilization of residual optical signals beneath them, and thick cloud regions are reconstructed using image inpainting techniques to eliminate the impact of thick cloud on the result.

2) We introduce the CFTM, ensuring the coherence between cloud removal result in the cloud regions and cloud-free regions. The CFTM is capable of capturing features at different scales, enabling it to recover large-scale image structures and capture small-scale details. In particular, we use the global structural information provided by SAR to compute the affinity between the cloud-free regions and the cloud regions in CFTM, which makes the confidence of our computed affinity higher.
3) We adopt the adversarial loss for high-resolution bands and apply the reconstruction loss and the multiscale structural similarity (MS-SSIM) loss for all the bands. The adversarial loss enhances high-frequency details in the high-resolution bands. The reconstruction loss reduces the gap between the results and the real cloud-free images, and the MS-SSIM loss optimizes the results based on the human visual system.

The rest of this article is organized as follows. Section II introduces the related works on cloud removal. The details of the proposed scheme are given in Section III. Section IV lists extensive experiments to evaluate the effectiveness of our proposed scheme. In Section V, we conclude this article and discuss future work.

II. Related Works

Cloud coverage can lead to the missing of remote sensing optical data, which is not conducive to remote sensing applications. In recent years, various methods are proposed for the missing information reconstruction of remote sensing data that can be used for cloud removal. Traditional methods can be categorized into three major categories: spatial-based methods, spectral-based methods, and temporal-based methods.

The spatial-based methods draw on the idea of image inpainting. It is assumed that the cloud-free regions and the cloud regions have similar texture structures. Information from the cloud-free regions is used to synthesize the information for cloud regions, and no other auxiliary information sources are used. Meng et al. [23] adaptively reconstructed thick cloud regions using a sparse dictionary learning algorithm. Cheng et al. [24] proposed a method based on multichannel nonlocal total variation for recovering missing regions in remote sensing images and achieved spectral coherence for the recovery result. Maalouf et al. [25] reconstructed cloud regions by propagating the image's geometrical flow curves obtained using the Bandelet transform with multiscale grouping. Spatial-based methods are effective when the cloud regions are small, but such methods cannot recover large thick cloud regions easily.

The spectral-based methods mainly exploit the relationship between different bands. Auxiliary information is extracted from other bands to recover the information in the band corrupted by clouds. In multispectral data, different bands have different abilities to penetrate clouds. Some bands can penetrate clouds, and the information of these bands can be used to recover the information in the corrupted band. Rakwatin et al. [26] used histogram-matching and local least-squares fitting algorithms to recover the missing data of Aqua MODIS band 6. Xu et al. [27] detected and removed clouds by physical modeling and linear spectral mixture analysis. These methods are more effective for thin cloud removal, but they often fail when clouds reach a certain thickness where all the optical bands cannot penetrate.

For large thick cloud, temporal-based methods are proposed, which mainly integrate information from multiple observations of the same land at other adjacent times. The integrated information is used as auxiliary information to recover the cloud region information. Ji et al. [28] reconstructed the missing information in remote sensing data using the local low-rank tensor completion method and considering the relationship between three domains: spatial, spectral, and temporal. Li et al. [29] performed stepwise radiometric adjustment of the remote sensing images at adjacent times. The cloud regions in the target image are replaced with the corresponding regions of the adjusted adjacent-time images. Li et al. [30] fused images from another time point and two or more low-resolution images to serve as reference images, and then removed clouds from the optical image using a nonnegative matrix factorization method and improved the cloud removal results by an error correction method. Although temporal-based methods can solve the problem of large-area cloud coverage, they require that the land information should not change significantly between adjacent time points and that the same observation area cannot be covered by clouds frequently.

Compared with traditional cloud removal methods, deep learning methods in remote sensing have shown their powerful performance, which attracted researchers to integrate deep learning into the research of cloud removal. For remote sensing images covered by thin cloud, deep learning methods can directly remove the cloud using the optical information present under the thin cloud. For example, Li et al. [31] used the information of all the bands for thin cloud removal and designed a multiple-input and multiple-output architecture for the original resolution bands, avoiding the introduction of noise by rescaling different resolution bands to the same resolution. Xu et al. [32] used two networks, the model beneficiary network and the objective beneficiary network, to guide each other in the training process and jointly enhance the effectiveness of thin cloud removal. Xu et al. [33] presented a cloudy scene synthesis paradigm that produces cloudy images for arbitrary optical cloud-free observations and developed a multiscale generative adversarial network to remove thin cloud from cloudy images. In addition, the attention mechanism is also used in cloud removal tasks since it enables the network to pay more attention to the cloud regions in the cloudy image. Xu et al. [34] used the attentive recurrent network to generate the attention map of the cloudy image, which indicates the spatial distribution of clouds. Based on the attention map, the attentive residual

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network is used to generate the cloud-free feature maps, and then the reconstruction network reconstructs the cloud-free feature map into the cloud-free image. Pan [35] adopted a generative adversarial network and local-to-global spatial attention to perform cloud removal and recover information. For thick cloud removal, deep learning methods fuse multiple images from adjacent times to generate cloud-free images. For example, Sarukkai et al. [36] designed a multibranch spatiotemporal generator network. Images from different time points are input to different branches to learn the features and capture the correlation between multiple images. Finally, cloud-free images are generated by fusing the features of multiple images. Zhang et al. [37] proposed a spatio-temporal patch group recovery model to reconstruct patches of cloud-contaminated images. The recovery results of patches with overlapping regions are weighted and aggregated, and then the information of the whole cloud region is recovered by an iterative process. However, these methods are incapable of removing both thin and thick clouds that coexist in optical images, nor do they specifically target the removal of thin and thick clouds based on their distinct features.

For the simultaneous existence of thin and thick clouds in optical images, Chen et al. [38] considered all the cloud pixels and proposed an IHOT cloud correction method based on fitting trajectory to remove both thin and thick clouds. Hu et al. [39] used the support vector value contourlet transform to realize the decomposition of the thin cloud region and removed thin cloud through the processing of the high-frequency and low-frequency coefficients, and then removed thick cloud using a multioutput support vector regression learning method. However, these methods are difficult to address the issue of optical images being frequently covered by extensive clouds.

Recently, cloud removal methods combining deep learning and SAR images have been proposed, due to the fact that using SAR images of the same scene as auxiliary information can be beneficial for the information reconstruction of cloud regions. For example, Meraner et al. [13] concatenated SAR image with optical image and used a deep residual network architecture to remove cloud from remote sensing optical image. Considering the advantage of GAN in image translation tasks, some researches converted SAR images into optical images for cloud removal. For example, Darbaghshahi et al. [40] first used a generative adversarial network to convert the SAR image into an optical image, and then used another generative adversarial network and the transformed optical image to perform cloud removal on the cloud-contaminated image. Gao et al. [41] first translated SAR images into simulated optical images. Then GAN was used to fuse the SAR image, the simulated optical image, and the cloud-contaminated image to achieve the purpose of cloud removal. However, these methods that convert SAR images and cloudy images into cloud-free images may introduce contaminated cloud information into the network, leading to incomplete cloud removal or blurring phenomena.

III. Method

A. Overview of the Proposed Framework

The overall framework of the proposed model is shown in Fig. 2. It is a two-stage network to recover information in cloud regions, focusing on processing thin cloud and thick cloud separately and merging features from SAR images to reconstruct the cloud regions. The amount of information obscured by cloud cover varies with the thickness of the cloud. Thin cloud regions permit observation of ground objects, providing optical information for the cloud removal task. In contrast, such surface information is completely missing in thick cloud regions. Our research is capable of generating a corrections map describing the thickness of the cloud in gray scale in the first stage, and then converting this corrections map into a binary mask M based on a preset threshold to distinguish thick cloud regions and other regions. Thick cloud regions are then further processed in the second stage. These two stages constitute our framework. Given Sentinel-1 SAR image $S1$ and Sentinel-2 optical image $S2$, the first stage uses a residual network structure with $S1$ and $S2$ as inputs to remove thin cloud and predict the mask M for the thick cloud regions. The result of thin cloud removal is denoted as $\widehat{S2}$. In the second stage, a generator takes $S1$, $\widehat{S2}$, and M as inputs to reconstruct information in thick cloud regions. The CFTM is used to fill thick cloud regions with features borrowed from distant spatial locations, ensuring the consistency of global structure. In addition, optical and SAR images are input into the network using the concatenated manner.

B. Thin Cloud Removal Module

Inspired by the residual network architecture [19], we design the thin cloud removal module using dilated convolution [42] and gated convolution [43]. Dilated convolution can obtain a larger receptive field without increasing the number of parameters and computational effort. As the spatial structures in remote sensing images are similar, a larger receptive field can capture more spatial information. Traditional convolution considers all the pixels in the image, including those in the cloud regions and the cloud-free regions, as valid pixels. This is useful for rectangular regions to be reconstructed. However, for free-form cloud regions, it may cause visual defects surrounding cloud regions and produce edge responses. For cloudy images, pixels in thick cloud regions are meaningless, pixels in thin cloud regions refract some of the optical information, and pixels in cloud-free regions are the most useful for the results. In contrast, gated convolution adopts the learnable soft mask update rule for dynamic feature selection of feature maps, making it better suited for the irregular shapes of cloud regions. For each pixel at position (x, y) of the input I, a gated convolution operation is performed on it as follows:

$$F_{(x,y)} = \sum_{i=-\frac{k}{2}}^{\frac{k}{2}} \sum_{j=-\frac{k}{2}}^{\frac{k}{2}} W_{i,j}^{f} \cdot I_{x+i,y+j} \tag{1}$$

$$G_{(x,y)} = \sum_{i=-\frac{k}{2}}^{\frac{k}{2}} \sum_{j=-\frac{k}{2}}^{\frac{k}{2}} W_{i,j}^{g} \cdot I_{x+i,y+j} \tag{2}$$

$$O_{(x,y)} = \phi\left(F_{(x,y)}\right) \odot \sigma\left(G_{(x,y)}\right) \tag{3}$$

where O denotes the output of convolution, and I denotes the input of convolution. W^f denotes the parameter of the convolution kernel that computes the feature maps, W^g denotes the

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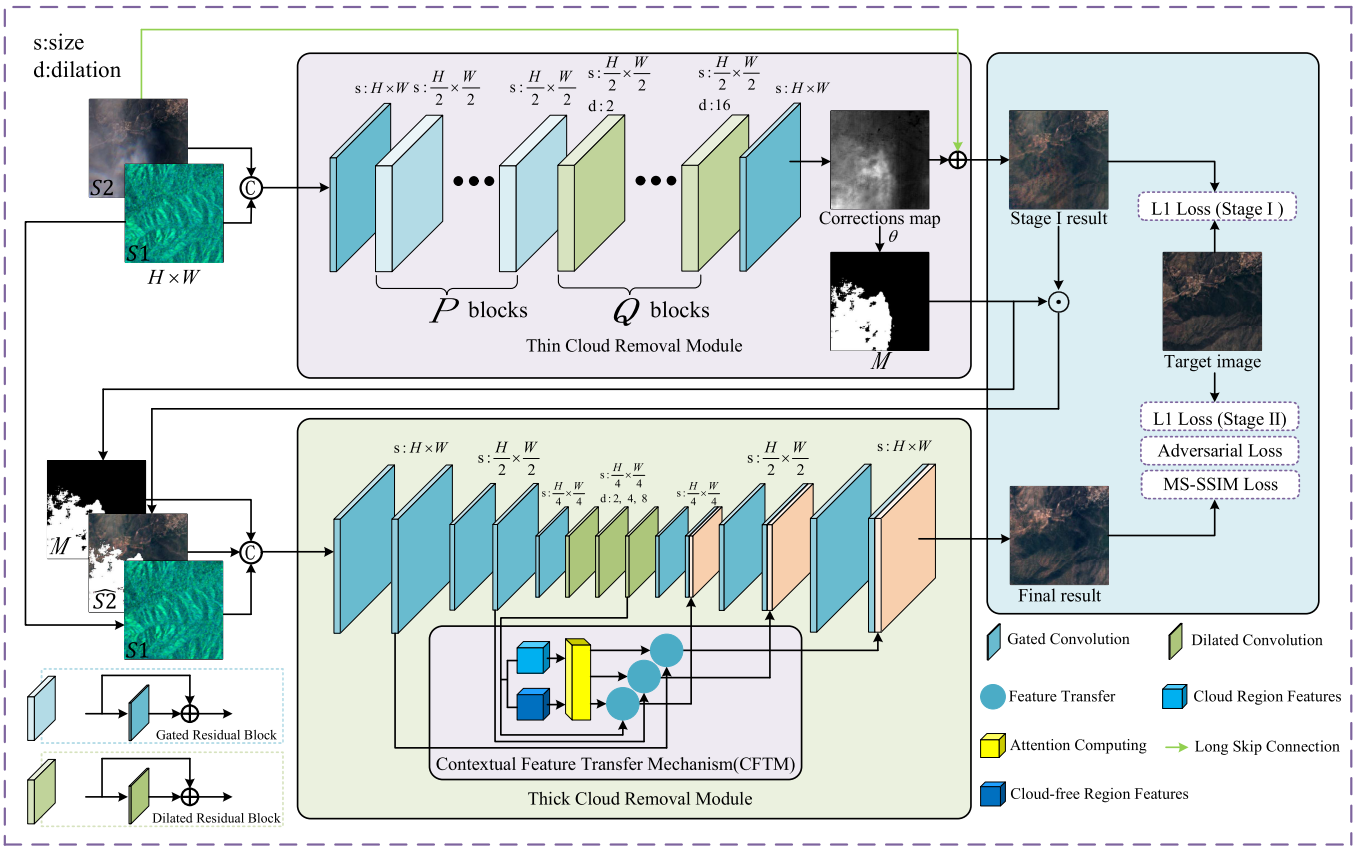


Fig. 2. Overall framework of the proposed method. It is a two-stage cloud removal network. A thin cloud removal network containing P gated residual blocks and Q dilated residual blocks is first performed to remove thin cloud and generate thick cloud mask. Then, a thick cloud removal network containing a CFTM is used to reconstruct information of the thick cloud regions.

parameter of the convolution kernel that computes the gating values, k denotes the size of the convolution kernel, ϕ denotes an activation function, and σ denotes sigmoid function, and thus the output value $\sigma(G_{(x,y)})$ is within [0, 1], as a soft gating mask. Gated convolution enables the learning of a dynamic feature selection mechanism for each spatial position. This mechanism effectively discards invalid pixels in thick cloud regions and reduces the weight of pixels in thin cloud regions.

The residual network architecture allows information to be transmitted directly to the output of the network, which is advantageous for cloud-free regions and thin cloud regions. For cloud-free regions, the output of the network only needs to reproduce the input, so the residual blocks are close to identity mapping, and the optical information of cloud-free regions can be transmitted to the output layer of the network. For thin cloud regions, shortcut connections can transmit optical information under thin cloud and learn a residual correction. The long skip connection plays an important role, which adds up the input optical remote sensing images and the output of the network. It means that the thin cloud removal network learns the corrections of the cloudy images, almost zero corrections for cloud-free regions and more corrections for cloud regions. Both the optical and SAR information can be regarded as references for thin cloud regions, but only SAR information is available for thick cloud regions. It leads to an unsatisfied result in thick regions, which need to be refined in the second stage.

The corrections map learned by the thin cloud removal network represents the thickness of the cloud, which can serve as a basis for identifying thick cloud regions. In this article, the optical image has 13 bands, and the network generates a corrections map for each band. The corrections map of Band 2 is used to measure the thickness of clouds, with values in the range $[-1, 0]$. The corrections map shows the spatial distribution of clouds, with larger opposite number of values indicating thicker clouds. The regions with thickness exceeding the threshold value θ are categorized as thick cloud regions. This operation can be expressed as

$$M_{(x,y)} = \begin{cases} 0, & -C_{(x,y)} < \theta \\ 1, & -C_{(x,y)} \geq \theta \end{cases} \tag{4}$$

where $C_{(x,y)}$ denotes the correction of the pixel at position (x, y) in the cloudy image, its value is negative, and its opposite represents the thickness of the cloud. The thick cloud mask M is a binary matrix where 1 denotes thick cloud regions, which correspond to regions to be reconstructed in the second stage.

C. Thick Cloud Removal Module

The thick cloud removal module adopts an encoder–decoder network and CFTM, which is shown in Fig. 3. The motivation behind incorporating CFTM stems from the inherent self-similarity observed in land regions of remote sensing

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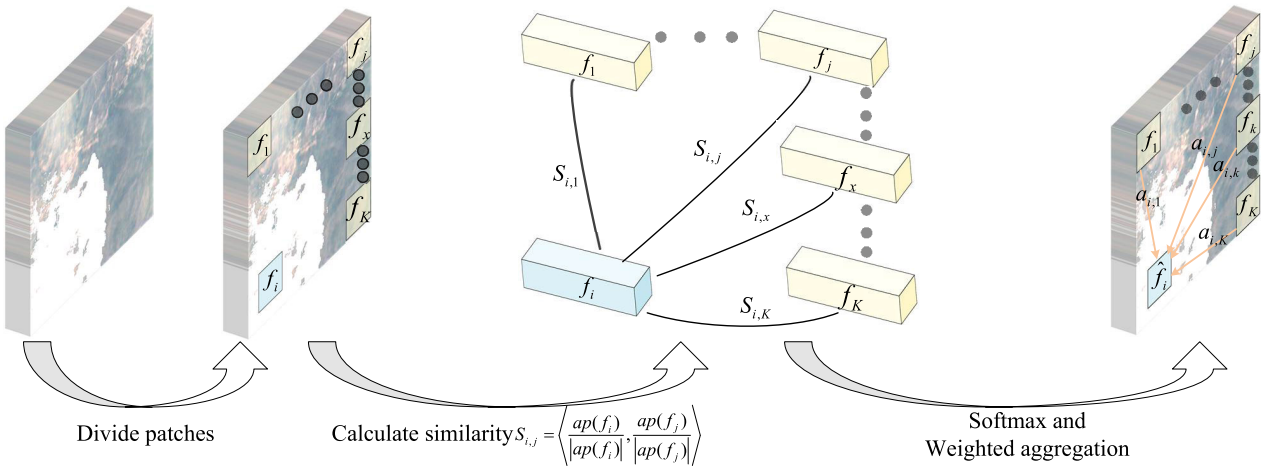


Fig. 3. Operational flowchart of the CFTM.

images. The convolution can only capture local information while neglecting other contextual information. The CFTM allows for the acquisition of information across a global scope. This mechanism borrows features from distant spatial locations to fill cloud regions, thereby maintaining the similarity between the reconstruction results in cloud regions and cloud-free regions. In addition, we use CFTM at multilevel feature maps, which can retain richer information such as edge details and textures.

The specific operation of CFTM is described as below. The feature map is divided into 32 × 32 patches. These patches are distinguished into cloud regions patches and cloud-free regions patches according to the thick cloud mask, and the cosine similarity between the two types of patches is calculated

$$s_{i,j} = \left\langle \frac{\text{ap}(f_i)}{|\text{ap}(f_i)|}, \frac{\text{ap}(f_j)}{|\text{ap}(f_j)|} \right\rangle \tag{5}$$

where ap(∗) denotes the global average pooling operation on ∗, f_i is the ith patch extracted from feature map inside cloud regions, and f_j is the jth patch extracted from feature map inside cloud-free regions. Then softmax is applied on the similarities to obtain the attention score for each patch inside cloud regions

$$a_{i,j} = \text{Softmax}(s_{i,j}) = \frac{s_{i,j}}{\sum_j^K s_{i,j}} \tag{6}$$

where K denotes the number of patches inside cloud-free regions. Contextual cloud-free patches are weighted and aggregated by the attention score to fill the cloud regions in the feature map

$$f_i = \sum_j^K a_{i,j} f_j. \tag{7}$$

The CFTM is applied on multilevel feature maps, and the reconstructed feature maps generated by the CFTM are transferred to the corresponding level of the decoder through a skip connection.

D. Loss Functions

Loss functions used in this article can be divided into three terms: the reconstruction loss, the MS-SSIM loss [44], and the adversarial loss. We apply the reconstruction loss and the MS-SSIM loss to all 13 bands. As bands 2/3/4/8 (visible and near-infrared, VNIR) have the highest resolution, we only apply the adversarial loss to these four bands.

The reconstruction loss is the L1 distance between cloud removal result and the real cloud-free image, which can be used to recover pixel-level information. The reconstruction loss is calculated as follows:

$$L_1 = \|x - G(z)\|_1^1 \tag{8}$$

where x is the cloud-free image and z is the input cloudy image. $G(\cdot)$ denotes the model output.

The MS-SSIM loss measures the structural similarity between cloud removal result and the real cloud-free image at different scales. The MS-SSIM loss can be formulated as

$$L_{\text{MS-SSIM}} = 1 - \text{MS-SSIM}(x, G(z)). \tag{9}$$

The adversarial loss can enable the network to capture high-frequency details and generate more realistic images. Rescaling multiresolution bands of the input image to the same resolution can result in low-resolution bands becoming blurry, thus the adversarial loss is applied only in the high-resolution bands. The adversarial loss is calculated as follows:

$$\begin{aligned} L_{\text{GAN}} = \min_G \max_D & E_{x_r \sim p_r}\left[\log(D(x_r))\right] \\ & + E_{x_c \sim p_c}\left[\log(1 - D(G(x_c)))\right] \end{aligned} \tag{10}$$

where p_r denotes the true cloud-free data distribution and p_c denotes the cloudy data distribution. $D(\cdot)$ is the discriminator output, which indicates the probability of judging the picture as true. $G(\cdot)$ is the model output.

In the first stage, only reconstruction loss L_1 is used. The total loss for the second stage is formulated as

$$L = L_1 + \lambda_1 L_{\text{MS-SSIM}} + \lambda_2 L_{\text{GAN}} \tag{11}$$

where λ_1 and λ_2 denote the hyperparameters used to balance losses.

IV. Experimental Validation

A. Datasets and Evaluation Measurements

In this article, we use the public dataset SEN12MS-CR [45], which contains 169 nonoverlapping regions of interest (ROIs),

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149 of which are used for training, ten for validation, and ten for testing. The average size of each ROI is about 52×40 km^2 ground coverage, which corresponds to an image of about 5200×4000 pixels. Each ROI generates on average more than 700 patches of 256×256 pixels, with 50% spatial overlap between neighboring patches. Each data entry contains triplets of the optical cloudy image, the optical cloud-free image and the SAR image. The optical image contains 13 bands and the SAR image contains two polarization channels. In data preprocessing, this article uses the image size 256×256 for training and testing. The optical image is clipped to values $[0, 10\,000]$. For VV and VH polarizations of the SAR image, the values are clipped to $[-25, 0]$ and $[-32.5, 0]$, respectively. Then, the optical image and the SAR image are scaled to the range $[0, 1]$.

The proposed approach is implemented using Python language with the PyTorch library and trained on 4 NVIDIA GeForce RTX 2080 Ti GPUs. The weights of the network are initialized using a uniform He distribution [46]. The model is trained using Adam [47] optimizer, and the batch size is set to 4. The networks in both the stages are trained separately and the epochs of training iterations are set to 50. The learning rate starts at 10^{-4}, which decays by 50% every ten epochs. In this work, the number of gated residual block P is set to 4, and the number of dilated residual block Q is set to 4. The hyperparameter θ is set to 0.07, and the hyperparameters λ_1 and λ_2 in the loss function are set to 0.1 and 0.001, respectively.

To quantitatively evaluate the results of cloud removal, we adopt the following metrics to measure the results: the mean absolute error (MAE), the peak signal-to-noise ratio (PSNR), the spectral angle mapper (SAM) [48], and the structural similarity index (SSIM) [49], which are defined as follows:

$$\mathrm{MAE}(x, y) = \frac{1}{C \cdot H \cdot W} \sum_{c=h=w=1}^{C,H,W} \left| x_{c,h,w} - y_{c,h,w} \right| \tag{12}$$

$$\mathrm{PSNR}(x, y) = 10 \cdot \log_{10}\left(\frac{MAX^2}{\mathrm{MSE}(x, y)} \right) \tag{13}$$

$$\mathrm{SAM}(x, y) = \cos^{-1} \times \left(\frac{\sum_{c=h=w=1}^{C,H,W} x_{c,h,w} \cdot y_{c,h,w}}{\sqrt{\sum_{c=h=w=1}^{C,H,W} x_{c,h,w}^2 \cdot \sum_{c=h=w=1}^{C,H,W} y_{c,h,w}^2}} \right) \tag{14}$$

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + \epsilon_1\right)\left(2\sigma_{xy} + \epsilon_2\right)}{\left(\mu_x + \mu_y + \epsilon_1\right)\left(\sigma_x + \sigma_y + \epsilon_2\right)} \tag{15}$$

where x and y are the images to be compared; C is the number of channels; H and W are the size of images; MAX is the maximum value of any pixel in the images, and 0 is the mean squared error between the two images; μ_x and μ_y are the means; σ_x and σ_y are the standard deviations; σ_{xy} is the covariance; and ϵ_1 and ϵ_2 are default constants.

B. Comparisons With Other Models

To fully evaluate the performance of our proposed model, we adopt the same input size of 128×128 as that of other models during training and testing phases for conducting comparative experiments. We compare our approach with Pix2pix [50], the two-stage cloud removal method using U-Net and GAN (UNet-GAN) [12], SF-GAN [41], SpA-GAN [35], DSen2-CR [13], and GLF-CR [14]. Pix2pix is an image transformation method that converts cloudy images and SAR images into cloud-free images. UNet-GAN is another two-stage method for cloud removal, but this method does not incorporate SAR images. To ensure fairness in our analysis, this article introduces two variants of UNet-GAN: one using SAR images (*w* SAR) and the other without using of SAR images (*w/o* SAR). SF-GAN uses two generators and divides the process into two steps to transform SAR images and cloudy images into cloud-free images. SpA-GAN uses a generator to transform optical cloudy images into cloud-free images. DSen-CR and GLF-CR are currently the most popular cloud removal methods.

TABLE I
QUANTITATIVE COMPARISONS OF THE RESULTS OF DIFFERENT METHODS

Method	MAE↓	SAM↓	PSNR↑	SSIM↑
SpA-GAN	0.0427	13.5344	24.8922	0.7694
UNet-GAN *w/o* SAR	0.0339	10.5707	26.9428	0.8397
Pix2pix	0.0312	9.9375	27.7904	0.8514
UNet-GAN *w* SAR	0.0313	9.6331	27.8001	0.8429
SF-GAN	0.0310	9.5719	27.9102	0.8654
DSen2-CR	0.0308	8.8627	28.0921	0.8751
GLF-CR	0.0276	8.3178	29.0410	0.8845
Ours	**0.0254**	**7.8019**	**29.6343**	**0.8862**

The quantitative results in Table I indicate that our method outperforms the existing methods in terms of MAE, SAM, PSNR and SSIM measures. Fig. 4 shows the overall comparison of the visualization results of different methods. Fig. 5 shows the local detail comparison of the different methods.

The visualization results indicate that our model outperforms other models. The Pix2pix model takes concatenated multispectral cloudy images and dual-polarimetric SAR images as input, and it converts them into cloud-free optical images through encoding and decoding. However, Pix2pix does not perform well in learning the mapping transformation relationship for large thick cloud regions, making it difficult to reconstruct clear ground information. The UNet-GAN without SAR images is unable to effectively recover information beneath extensive thick cloud regions, such as the second and third rows in Fig. 4. The UNet-GAN with SAR images can recover the structure of cloud regions but struggles to restore fine texture details beneath clouds. SF-GAN first transforms SAR images into simulated optical images and then converts SAR images, cloudy images, and simulated optical images into cloud-free images. However, the results exhibit blurriness, indicating a drawback of the conversion method. On the other hand, SpA-GAN only uses optical images as input. From the results, it can be observed that the removal of large-area thick cloud is a complete failure, confirming the significance of SAR images in effectively removing extensive thick cloud regions. Some images produced by DSen2-CR and GLF-CR do not completely remove clouds or are accompanied by blurring phenomena. The long skip connections in these two methods can effectively use the ground information in cloudy images,

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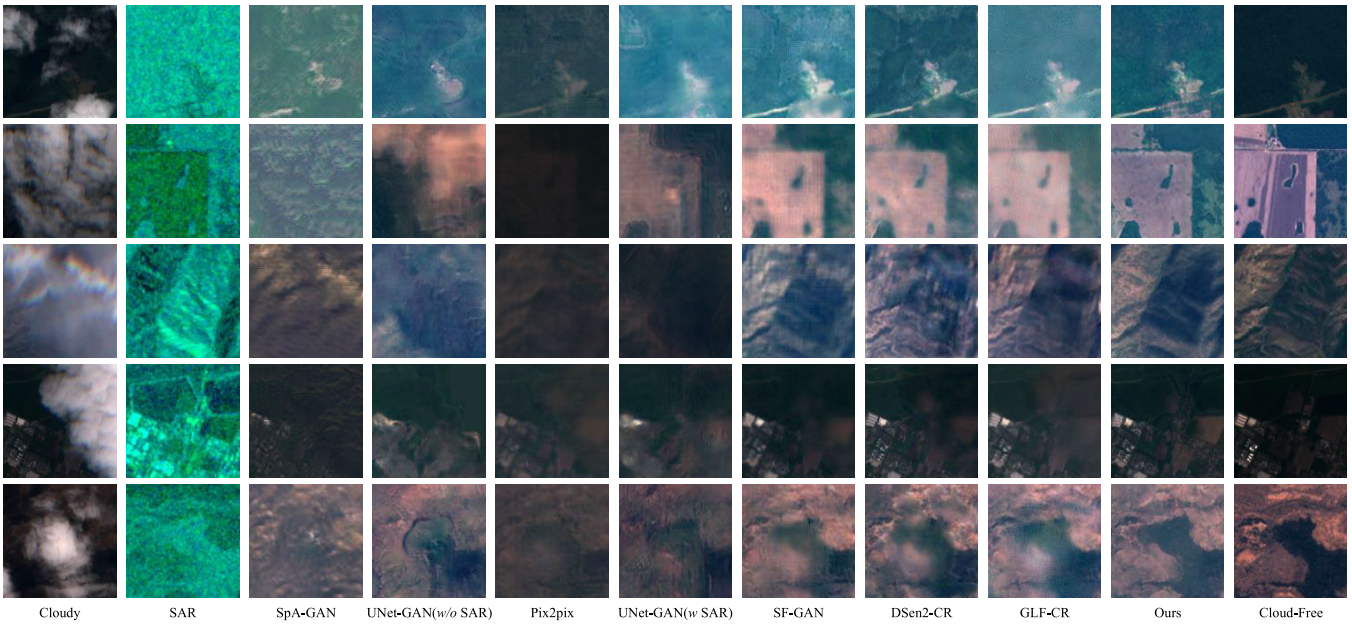


Fig. 4. Qualitative results of different cloud removal methods. For each row, from left to right are, respectively, the cloudy image, the SAR image, the result from SpA-GAN, UNet-GAN (*w/o* SAR), Pix2pix, UNet-GAN (*w* SAR), SF-GAN, DSen2-CR, GLF-CR, our model, and the target cloud-free image.

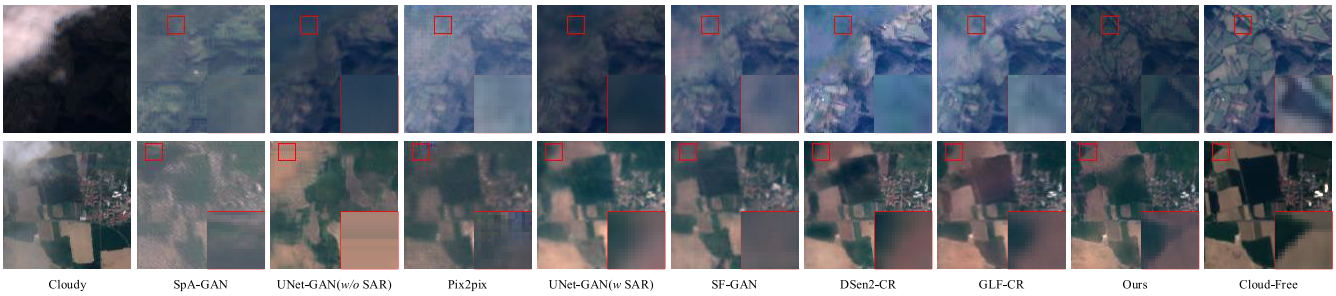


Fig. 5. Zoomed-in view of the local details of the result. Enlarged details are supplemented at the corner of each image.

TABLE II
QUANTITATIVE COMPARISON OF THE PERFORMANCE OF OUR METHOD USING DIFFERENT CLOUD REMOVAL STAGES

	MAE↓	SAM↓	PSNR↑	SSIM↑
Thin cloud removal stage	0.0293	8.5758	28.4065	0.8437
Thick cloud removal stage	0.0276	8.3292	28.8878	0.8792
Thin & Thick stage	**0.0254**	**7.8102**	**29.6133**	**0.8860**

but they also introduce visual defects, especially in thick cloud regions. Our method designs a two-stage cloud removal scheme, where the first stage uses the residual architecture and long skip connections to effectively preserve the ground information, and the second stage reconstructs thick cloud regions, mitigating the impact of a large number of cloud pixels on the results. In addition, using the CFTM on feature maps at different levels, features from cloud-free regions are used to fill in cloud regions. Consequently, our method produces fine results without blurring.

C. Importance of Two-Stage Cloud Removal

To avoid interference from clouds during the restoration of information, we have applied a masking process to the cloud layer. This means that all the information in cloud regions is considered invalid, but obviously, there is still optical information under the thin cloud. Masking thin cloud would result in a waste of information during reconstruction. Therefore, we design a module for removing thin cloud before removing thick cloud to make full use of the optical information underneath thin cloud. The proposed two-stage cloud removal scheme, which first removes thin cloud and then thick cloud, fully considers the distinct features of both types of clouds and designs different network structures for each. To assess the effectiveness of the two stages, we compared the results of testing with only the thin cloud removal stage (CR1), testing with only the thick cloud removal stage (CR2), and testing with the full model. Additional cloud masks are required for testing with CR2. In this experiment, the cloud detection method proposed by Schmitt et al. [51] was used to calculate the cloud mask.

Fig. 6 shows the visualization results of CR1, CR2, and the full model, verifying the effectiveness of two-stage cloud removal. The green ellipse marks the restoration result of thin cloud regions. As shown in Fig. 6, CR1 and full model can restore fine-grained details, whereas CR2 can only recover

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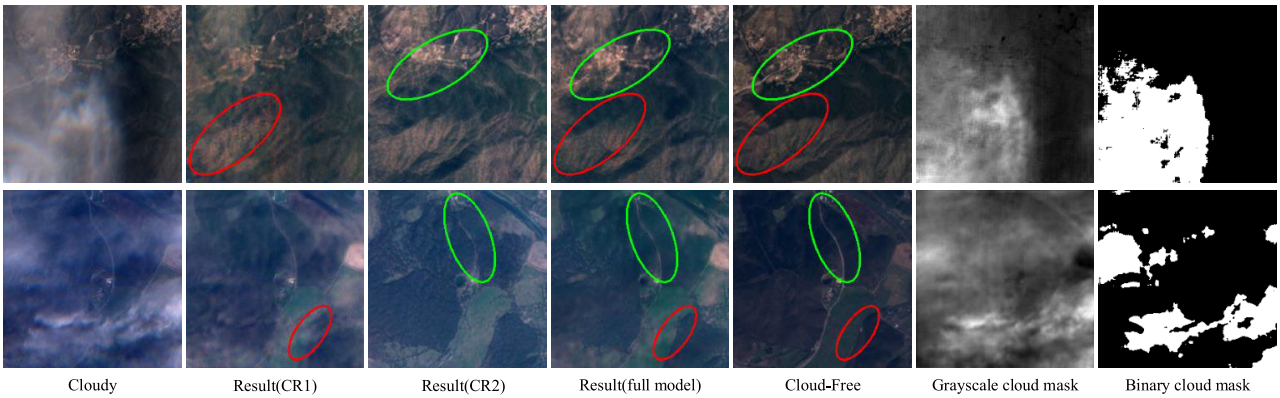


Fig. 6. Results of the two-stage ablation experiment. For each row, from left to right are, respectively, the cloudy image, restoration result using only the thin cloud removal stage (CR1), restoration result using only the thick cloud removal stage (CR2), restoration result using full model, the target cloud-free image, corrections map (i.e., grayscale cloud mask) generated during the thin cloud removal stage, and the binary cloud mask. The grayscale cloud mask is a single-band image which shows the thickness of the cloud. The binary cloud mask delineates the thick cloud regions to be restored in the second stage.

information under thin cloud on a coarse scale. This disparity in results is due to the fact that CR1 and full model use not only SAR images but also the optical information to restore thin cloud regions. CR2 considers all the cloud regions as invalid information, meaning that restoration of the information under thin cloud can only rely on SAR images, without any optical information. Due to the heterogeneity between SAR images and optical images, SAR images cannot provide optical details as finely as optical images do. Therefore, CR2 fails to use the optical information beneath thin cloud, resulting in the coarse recovery of information.

The red ellipse marks the restoration results of thick cloud regions. CR1 can cause color distortion and blurring when removing thick cloud. This is because CR1 learns the correction for each pixel in the optical cloudy image. For thick cloud regions, too large corrections can lead to color distortion, insufficient corrections can result in incomplete cloud removal, and similar corrections can produce blurring. However, for thin cloud regions, only small corrections are needed to remove the thin cloud, and similar corrections will not cause blurring due to the presence of optical information. The restoration effect of CR1 on thick cloud regions is not as good as that on thin cloud regions. Therefore, the image after removing thin cloud needs to be input into the thick removal stage to reconstruct thick cloud regions, so that the final cloud removal result not only contains optical information under thin cloud but also ensures that it is not affected by thick cloud. Table II shows the quantitative results.

However, from the second row of Fig. 6, it can be observed that the results suffer from spectral distortions. This is due to the dataset containing 169 ROIs (i.e., 169 classes), with inconsistent spectral characteristics among classes and an uneven distribution of data. The spectral characteristics of multiple classes are presented in green, and the cloud-free regions in the cloudy image of the second row also appear green. Consequently, the network prediction result was biased toward green hues, resulting in spectral distortions.

D. Importance of SAR Images

To verify the importance of SAR images as auxiliary information that can provide structural information, we design two modes of network training. All the training details of them are consistent, except that one mode incorporates SAR images while the other does not. The quantitative results from Table III indicate that incorporating SAR images as auxiliary information for cloud removal outperforms the use of optical images alone. The effectiveness of SAR images can also be observed in the visualization results of testing two modes, as shown in Fig. 7. These results highlight the impact of SAR images on improving the performance of cloud removal methods.

From Fig. 7, it can be seen that the results reconstructed using SAR images as auxiliary information contain more structural information in comparison, particularly in the reconstruction of large cloud regions. For small cloud regions, the information of cloudy regions can be predicted using information from cloud-free regions, such as the first row of

TABLE III
QUANTITATIVE EVALUATION OF SAR IMPORTANCE IN OPTICAL IMAGE CLOUD REMOVAL

	MAE↓	SAM↓	PSNR↑	SSIM↑
HF-T2CR *w/o* SAR	0.0384	11.1439	25.8291	0.8316
HF-T2CR	**0.0254**	**7.8102**	**29.6133**	**0.8860**

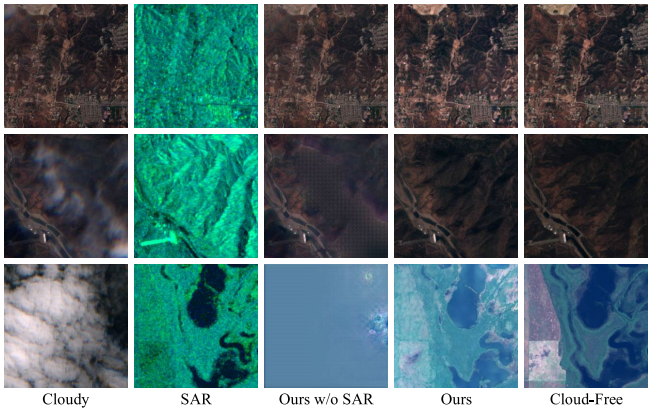


Fig. 7. Qualitative study on the importance of SAR images. For each row, from left to right are, respectively, the cloudy image, the SAR image, the result without the SAR input, the result including the SAR input, and the target cloud-free image.

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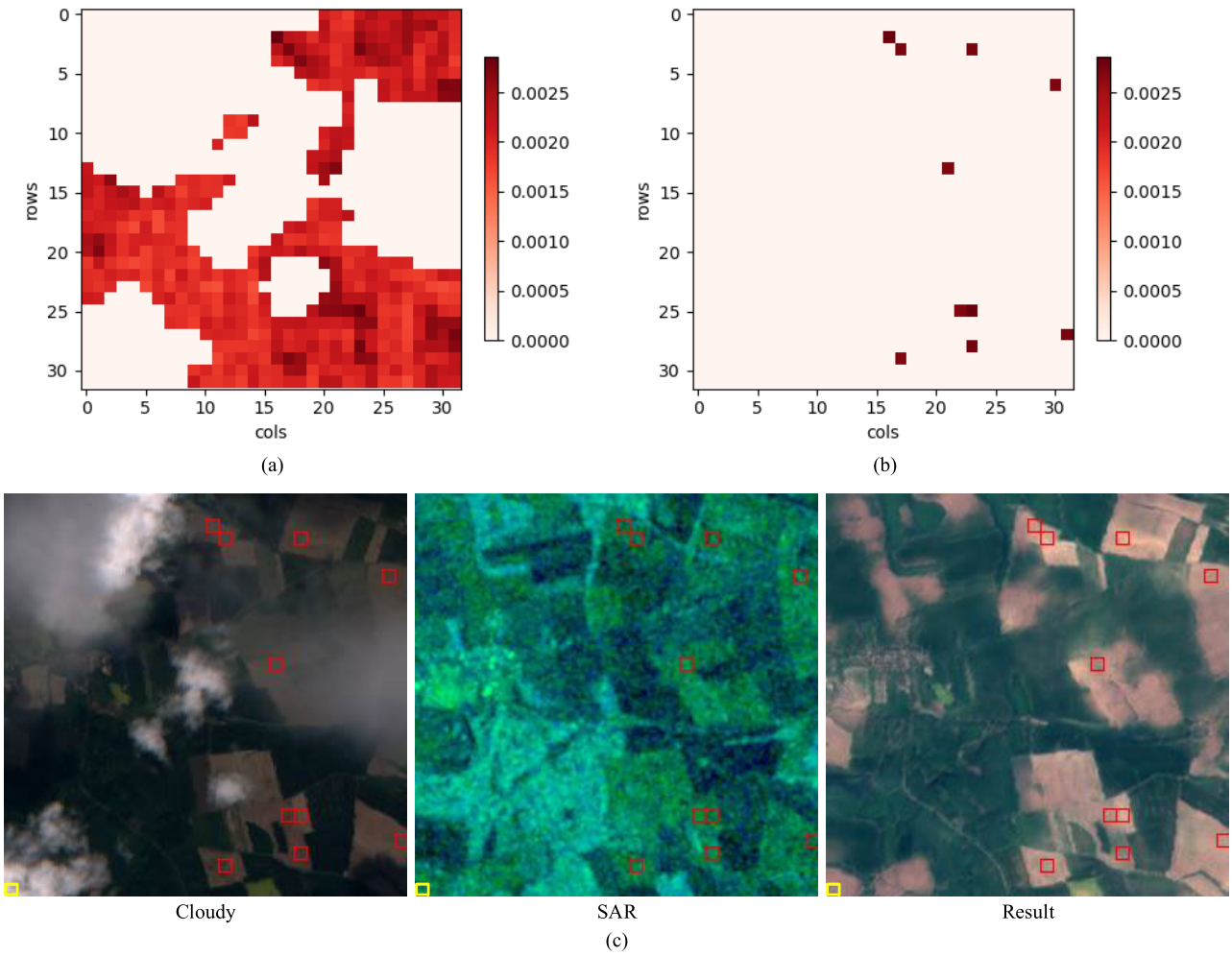


Fig. 8. Visualization of the CFTM. (a) Attention map of the yellow-boxed region in (c). (b) Visualizes the top ten attention scores. (c) Top ten regions in the optical and SAR images that are most similar to the yellow-boxed region, indicated by the red-boxed regions. The cloud removal result of the yellow-boxed region and the land type within red-boxed regions are consistent.

Fig. 7, even without the SAR image, cloud regions can be reconstructed using information from surrounding cloud-free regions. The reconstructed results are similar to the textures of the surrounding cloud-free regions. However, for large cloud regions, reconstruction fails without the assistance of SAR images, as shown in the second and third rows of Fig. 7.

This further indicates that in situations where features similar to the land cover type of areas obscured by clouds are few or nonexistent, CFTM cannot effectively use features from cloud-free regions for the recovery of information in cloud regions without the collaborative support of SAR images. Nevertheless, due to the presence of SAR images, the model is still capable of restoring the ground information of cloud regions, even when CFTM is unable to effectively use the features of cloud-free regions.

SAR images can provide edge structure information for large structures or ground targets with clear edges. With the help of SAR, our method is capable of recovering the structural information of these large-scale targets beneath the cloud cover. However, predicting specific small targets through SAR images can be challenging.

E. Visualization of CFTM

To better understand CFTM, we visualized the attention scores in this mechanism as shown in Fig. 8. In CFTM, the feature map is divided into 32×32 patches, with each patch representing an 8×8 pixels area of the input image. Fig. 8(a) shows the attention scores of the area within the yellow-boxed region located in the lower left corner. The attention map represents the similarity between the cloud-free regions and the yellow-boxed region. Fig. 8(b) indicates the top ten patches in the attention scores, and the red-boxed regions in Fig. 8(c) show the locations of these ten patches in the optical image. Since cloud regions obscure the optical information, the attention score relies more on information provided by SAR images for calculation. In the SAR image, it is observed that the red-boxed and yellow-boxed regions belong to the same land type. Thus, more attention is allocated to the red-boxed regions, and consequently, the yellow-boxed region is reconstructed with the same land type as red-boxed regions.

The visualization experiment demonstrates that CFTM can model long-range dependencies, using texture and structure

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TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON CFTM

	MAE↓	SAM↓	PSNR↑	SSIM↑
HF-T2CR *w/o* CFTM	0.0264	8.1287	29.0006	0.8811
HF-T2CR(CFTM on one level)	0.0256	7.8477	29.4968	0.8854
HF-T2CR(CFTM on two levels)	**0.0254**	7.8306	**29.6516**	0.8857
HF-T2CR(CFTM on three levels)	**0.0254**	**7.8102**	29.6133	**0.8860**

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON LOSS FUNCTION

	MAE↓	SAM↓	PSNR↑	SSIM↑
HF-T2CR(L_1)	0.0267	8.2891	29.0582	0.8807
HF-T2CR(L_1+$L_{\text{MS-SSIM}}$)	0.0257	7.8821	29.5084	0.8858
HF-T2CR(L_1+L_{GAN})	0.0263	8.1859	29.2661	0.8801
HF-T2CR(L_1+$L_{\text{MS-SSIM}}$+L_{GAN})	**0.0254**	**7.8102**	**29.6133**	**0.8860**

information from distant locations to synthesize land information in cloud regions.

F. Ablation Experiments of CFTM

CFTM fills the features of cloud-free regions into cloud-covered regions based on similarity weights. As this filling process is performed on three levels of feature maps, both visual and semantic coherence for cloud removal can be ensured. To verify CFTM, we conduct some ablation experiments, analyzing the impact of CFTM and its use at different levels. The new comparative models are: 1) without using CFTM; 2) using CFTM on one level only; and 3) using CFTM on two levels only, while other parts of the comparative models are the same as for our model. The results of the ablation experiments are shown in Table IV.

The results indicate that model using CFTM outperforms the model without it, especially with a decrease of 3.78% in MAE and a decrease of 3.91% in SAM. This is attributed to the ability of CFTM to compensate for the limitations of convolution in capturing only local information. CFTM enables the explicit utilization of features from cloud-free regions as references, guiding the model to make better predictions. CFTM works best on three levels feature maps, and using multiple levels is advantageous for information recovery of cloud regions at both the high-level semantic and texture feature levels.

G. Ablation Experiments of Loss Function

To further investigate the impact of the loss function on the results, we conducted ablation experiments of loss function. During experiments, distinct training loss were used: using only L_1, combining L_1 with $L_{\text{MS-SSIM}}$, and combining L_1 with L_{GAN}. The results of the ablation experiment are shown in Table V.

The results indicate a notable enhancement across all the four metrics upon the inclusion of $L_{\text{MS-SSIM}}$. This improvement is attributed to the ability of $L_{\text{MS-SSIM}}$ to evaluate structural disparities between the results and cloud-free images across multiple scales, allowing for a more comprehensive consideration of the structural, textural, and detailed characteristics. It enhances the quality of generated images and aligns more closely with human perceptual criteria. L_1 and $L_{\text{MS-SSIM}}$ individually fall short in capturing high-frequency details and in the authenticity and naturalness of the images. The L_{GAN} can alleviate this problem and it help the results to be closer to the distribution of the real image. The introduction of L_{GAN} yields a minor improvement in all the four metrics. The combination of L_{GAN} and $L_{\text{MS-SSIM}}$ has significantly optimized the model, leading to a 4.87% decrease in MAE and a 5.78% decrease in SAM.

V. CONCLUSION

Given the varying degrees of ground coverage by thin and thick clouds, with usable ground information available beneath thin cloud but entirely absent beneath thick cloud cover, it is challenging to remove both simultaneously. In this article, we propose a cloud removal scheme (HF-T2CR) that removes thin cloud first and then thick cloud, and uses SAR images as a guide, effectively addressing the difficult problem of cloud removal in complex cloud scenes. HF-T2CR can perform targeted cloud removal based on the distinct features of two cloud types, a capability that other methods lack. Initially, a network based on residual architecture is used to restore optical information beneath thin cloud regions and generate a grayscale cloud mask. Using a preset threshold on this grayscale cloud mask enables the distinction between thick cloud regions and other regions. And then an encoder–decoder network is used to reconstruct the information for thick cloud regions, eliminating the influence of thick cloud on the reconstruction results. Furthermore, a CFTM is used to fill thick cloud regions by borrowing features from distant locations. Benefiting from the two-stage cloud removal approach and CFTM, HF-T2CR preserves both optical information beneath thin cloud and ensures that the thick cloud regions in the results exhibit clear textures and visual and semantic consistency with other regions. In this aspect, our method significantly outperforms other models where the thick cloud regions are often blurry and susceptible to cloud interference. Extensive comparative and ablation experiments demonstrated the effectiveness of our method.

While our scheme yields satisfactory cloud removal results, we note that it adds additional computational overhead when processing optical images containing only thin or thick clouds. Future work will focus on using dynamic routing to process different cloud scenarios to reduce the extra computation.

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